



FACULTÉ DES SCIENCES ET DES TECHNOLOGIES(FST)

TD N°9_ – ADMIN-SYSTEME

COURS: ADMINISTRATION DES SYSTÈMES
INFORMATIQUES

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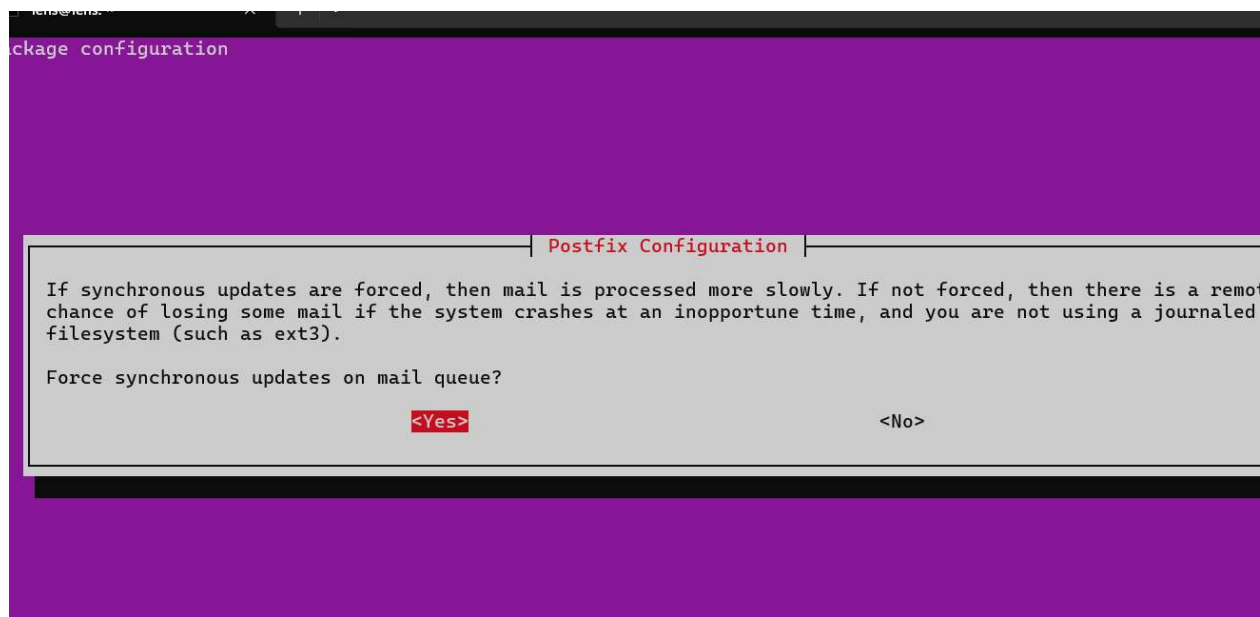
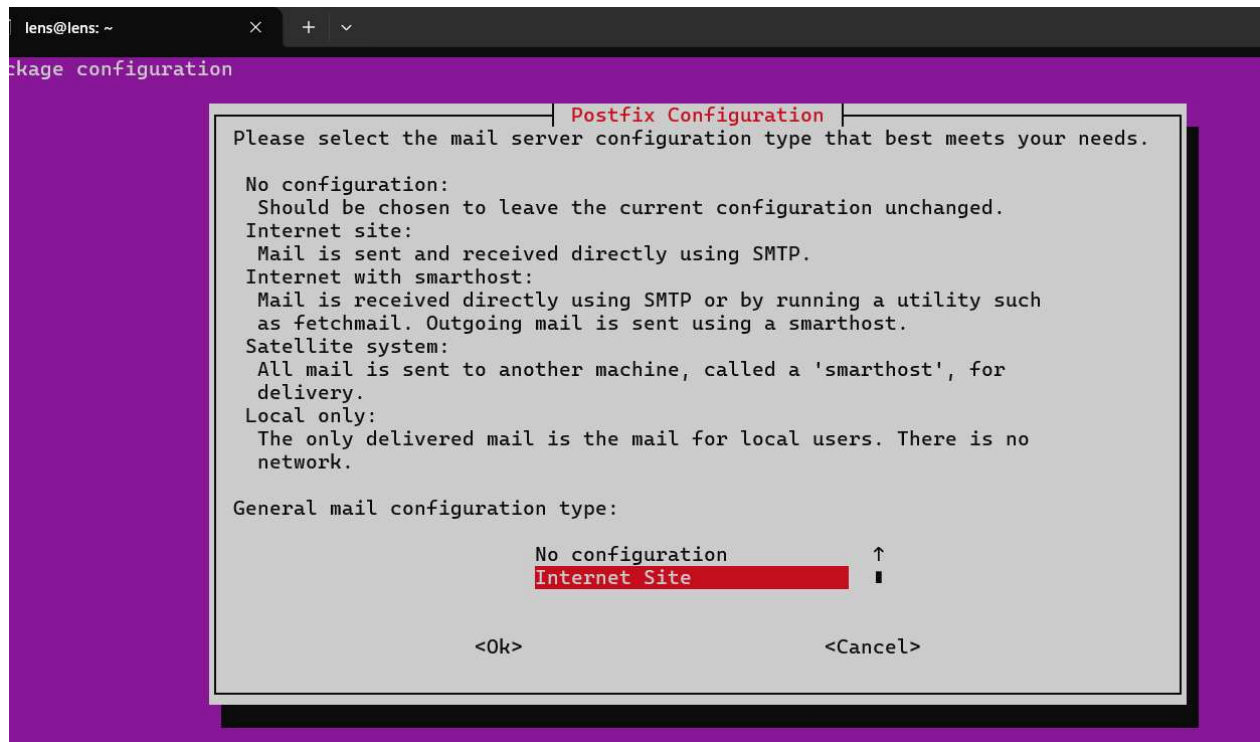
DATE: 29/04/2026

L'objectif de ce TD est de:

1. installer un serveur SMTP
2. configurer la réception IMAP / POP3
3. déployer un webmail professionnel
4. sécuriser les accès
5. analyser les logs
6. tester l'envoi / réception
7. documenter une architecture mail complete

Installation de Postfix (SMTP)

```
lens@lens:~$ sudo apt install postfix mailutils -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
postfix is already the newest version (3.8.6-1build2).
mailutils is already the newest version (1:3.17-1.1build3).
0 upgraded, 0 newly installed, 0 to remove and 79 not upgraded.
lens@lens:~$ |
```



Configuration

```
myhostname = mail.ius.ht
mydomain = ius.ht
myorigin = /etc/mailname
inet_interfaces = all
inet_protocols = ipv4
mydestination = ius.ht, mail.ius.ht, localhost.localdomain, localhost
mynetworks = 127.0.0.0/8, [::1]/128, 192.168.43.79/24
home_mailbox = Maildir/
|
```

Redémarrage

```
lens@lens:~$ sudo nano /etc/postfix/main.cf
lens@lens:~$ sudo systemctl restart postfix
lens@lens:~$ |
```

Installation de Dovecot (IMAP / POP3)

```
lens@lens: ~  
Creating config file /etc/dovecot/conf.d/auth-sql.conf.ext with new version  
Creating config file /etc/dovecot/conf.d/auth-static.conf.ext with new version  
Creating config file /etc/dovecot/conf.d/auth-system.conf.ext with new version  
Created symlink /etc/systemd/system/multi-user.target.wants/dovecot.service → /usr/lib/systemd/system/dovecot.service.  
dovecot.socket is a disabled or a static unit, not starting it.  
Setting up dovecot-imapd (1:2.3.21+dfsg1-2ubuntu6.4) ...  
  
Creating config file /etc/dovecot/conf.d/20-imap.conf with new version  
Setting up dovecot-pop3d (1:2.3.21+dfsg1-2ubuntu6.4) ...  
  
Creating config file /etc/dovecot/conf.d/20-pop3.conf with new version  
Processing triggers for ufw (0.36.2-6) ...  
Processing triggers for man-db (2.12.0-4build2) ...  
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...  
Processing triggers for dovecot-core (1:2.3.21+dfsg1-2ubuntu6.4) ...  
Scanning processes...  
Scanning linux images...  
  
Running kernel seems to be up-to-date.  
  
No services need to be restarted.  
  
No containers need to be restarted.  
  
No user sessions are running outdated binaries.  
  
No VM guests are running outdated hypervisor (qemu) binaries on this host.  
lens@lens:~$
```

Configuration

```
# Settings to control adding $HasAttachment or $HasNoAttachment keywords.  
# By default, all MIME parts with Content-Disposition=attachment, or inli  
# with filename parameter are consired attachments.  
#   add-flags - Add the keywords when saving new mails or when fetching c  
#   do it efficiently.  
#   content-type=type or !type - Include/exclude content type. Excluding  
#   never consider the matched MIME part as attachment. Including will  
#   negate an exclusion (e.g. content-type=!foo/* content-type=foo/bar)  
#   exclude-inlined - Exclude any Content-Disposition=inline MIME part.  
#mail_attachment_detection_options =  
mail_driver = maildir  
mail_path = ~/Maildir|
```

Protocoles

```
# Most of the actual configuration gets included below. The filenames
# first sorted by their ASCII value and parsed in that order. The 0
# in filenames are intended to make it easier to understand the order
!include conf.d/*.conf

# A config file can also be tried to be included without giving an error
# if it's not found:
!include_try local.conf
protocols = imap pop3
```

Redémarrage

```
lens@lens:~$ sudo systemctl restart dovecot
lens@lens:~$ sudo systemctl status dovecot
● dovecot.service - Dovecot IMAP/POP3 email server
   Loaded: loaded (/usr/lib/systemd/system/dovecot.service; enabled; pres>
   Active: active (running) since Fri 2026-05-01 18:19:23 UTC; 25s ago
     Docs: man:dovecot(1)
           https://doc.dovecot.org/
  Main PID: 5341 (dovecot)
    Status: "v2.3.21 (47349e2482) running"
     Tasks: 4 (limit: 2263)
    Memory: 3.4M (peak: 3.5M)
       CPU: 157ms
    CGroup: /system.slice/dovecot.service
            └─5341 /usr/sbin/dovecot -F
               └─5343 dovecot/anvil
                  └─5344 dovecot/log
                     └─5345 dovecot/config

May 01 18:19:23 lens systemd[1]: Starting dovecot.service - Dovecot IMAP/PO>
May 01 18:19:23 lens dovecot[5341]: master: Dovecot v2.3.21 (47349e2482) st>
May 01 18:19:23 lens systemd[1]: Started dovecot.service - Dovecot IMAP/POP>
lines 1-19/19 (END)
```

Création des comptes utilisateurs


```
postfix:x:111:111::/var/spool/postfix:/usr/sbin/nologin
nagios:x:112:113::/var/lib/nagios:/usr/sbin/nologin
zabbix:x:113:114::/var/lib/zabbix:/usr/sbin/nologin
Debian-snmp:x:114:115::/var/lib/snmp:/bin/false
_aide:x:988:988:Advanced Intrusion Detection Environment:/var/lib/aide:/usr/sbin/nologin
mysql:x:115:116:MySQL Server,,,:/nonexistent:/bin/false
prometheus:x:1003:1003::/home/prometheus:/bin/false
node_exporter:x:1004:1004::/home/node_exporter:/bin/false
grafana:x:116:117::/usr/share/grafana:/bin/false
etudiant1:x:1005:1005:,,,:/home/etudiant1:/bin/bash
etudiant2:x:1006:1006:,,,:/home/etudiant2:/bin/bash
etudiant3:x:1007:1007:,,,:/home/etudiant3:/bin/bash
etudiant4:x:1008:1008:,,,:/home/etudiant4:/bin/bash
etudiant5:x:1009:1009:,,,:/home/etudiant5:/bin/bash
etudiant6:x:1010:1010:,,,:/home/etudiant6:/bin/bash
etudiant7:x:1011:1011:,,,:/home/etudiant7:/bin/bash
etudiant8:x:1012:1012:,,,:/home/etudiant8:/bin/bash
etudiant9:x:1013:1013:,,,:/home/etudiant9:/bin/bash
etudiant10:x:1014:1014:,,,:/home/etudiant10:/bin/bash
etudiant11:x:1015:1015:,,,:/home/etudiant11:/bin/bash
etudiant12:x:1016:1016:,,,:/home/etudiant12:/bin/bash
etudiant13:x:1017:1017:,,,:/home/etudiant13:/bin/bash
etudiant14:x:1018:1018:,,,:/home/etudiant14:/bin/bash
etudiant15:x:1019:1019:,,,:/home/etudiant15:/bin/bash
etudiant16:x:1020:1020:,,,:/home/etudiant16:/bin/bash
etudiant17:x:1021:1021:,,,:/home/etudiant17:/bin/bash
etudiant18:x:1022:1022:,,,:/home/etudiant18:/bin/bash
etudiant19:x:1023:1023:,,,:/home/etudiant19:/bin/bash
etudiant20:x:1024:1024:,,,:/home/etudiant20:/bin/bash
etudiant21:x:1025:1025:,,,:/home/etudiant21:/bin/bash
etudiant22:x:1026:1026:,,,:/home/etudiant22:/bin/bash
etudiant23:x:1027:1027:,,,:/home/etudiant23:/bin/bash
etudiant24:x:1028:1028:,,,:/home/etudiant24:/bin/bash
etudiant25:x:1029:1029:,,,:/home/etudiant25:/bin/bash
prof1:x:1030:1030:,,,:/home/prof1:/bin/bash
prof2:x:1031:1031:,,,:/home/prof2:/bin/bash
prof3:x:1032:1032:,,,:/home/prof3:/bin/bash
prof4:x:1033:1033:,,,:/home/prof4:/bin/bash
prof5:x:1034:1034:,,,:/home/prof5:/bin/bash
prof6:x:1035:1035:,,,:/home/prof6:/bin/bash
prof7:x:1036:1036:,,,:/home/prof7:/bin/bash
prof8:x:1037:1037:,,,:/home/prof8:/bin/bash
prof9:x:1038:1038:,,,:/home/prof9:/bin/bash
prof10:x:1039:1039:,,,:/home/prof10:/bin/bash
admin1:x:1040:1040:,,,:/home/admin1:/bin/bash
admin2:x:1041:1041:,,,:/home/admin2:/bin/bash
admin3:x:1042:1042:,,,:/home/admin3:/bin/bash
admin4:x:1043:1043:,,,:/home/admin4:/bin/bash
admin5:x:1044:1044:,,,:/home/admin5:/bin/bash
dovecot:x:117:118:Dovecot mail server,,,:/usr/lib/dovecot:/usr/sbin/nologin
dovenull:x:118:119:Dovecot login user,,,:/nonexistent:/usr/sbin/nologin
etudiant26:x:1045:1045:etudiant etudiant,36465757,457586686,4747557676,0000:/home/etudiant26:/bin
Lens@Lens:/$ |
```

```
lens@lens: ~  
New password:  
Retype new password:  
passwd: password updated successfully  
Changing the user information for etudiant24  
Enter the new value, or press ENTER for the default  
    Full Name []:  
    Room Number []:  
    Work Phone []:  
    Home Phone []:  
    Other []:  
Is the information correct? [Y/n] y  
info: Adding new user 'etudiant24' to supplemental / extra groups 'users' ...  
info: Adding user 'etudiant24' to group 'users' ...  
info: Adding user 'etudiant25' ...  
info: Selecting UID/GID from range 1000 to 59999 ...  
info: Adding new group 'etudiant25' (1029) ...  
info: Adding new user 'etudiant25' (1029) with group 'etudiant25 (1029)' ...  
info: Creating home directory '/home/etudiant25' ...  
info: Copying files from '/etc/skel' ...  
New password:  
Retype new password:  
passwd: password updated successfully  
Changing the user information for etudiant25  
Enter the new value, or press ENTER for the default  
    Full Name []:  
    Room Number []:  
    Work Phone []:  
    Home Phone []:  
    Other []:  
Is the information correct? [Y/n] y  
info: Adding new user 'etudiant25' to supplemental / extra groups 'users' ...  
info: Adding user 'etudiant25' to group 'users' ...  
info: Adding user 'prof1' ...  
info: Selecting UID/GID from range 1000 to 59999 ...  
info: Adding new group 'prof1' (1030) ...  
info: Adding new user 'prof1' (1030) with group 'prof1 (1030)' ...  
info: Creating home directory '/home/prof1' ...  
info: Copying files from '/etc/skel' ...  
New password:  
Retype new password:  
passwd: password updated successfully
```



```
lens@lens: ~
Other []:
Is the information correct? [Y/n] y
info: Adding new user 'prof10' to supplemental / extra groups 'users' ...
info: Adding user 'prof10' to group 'users' ...
info: Adding user 'admin1' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group 'admin1' (1040) ...
info: Adding new user 'admin1' (1040) with group 'admin1 (1040)' ...
info: Creating home directory '/home/admin1' ...
info: Copying files from '/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for admin1
Enter the new value, or press ENTER for the default
Full Name []:
Room Number []:
Work Phone []:
Home Phone []:
Other []:
Is the information correct? [Y/n] y
info: Adding new user 'admin1' to supplemental / extra groups 'users' ...
info: Adding user 'admin1' to group 'users' ...
info: Adding user 'admin2' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group 'admin2' (1041) ...
info: Adding new user 'admin2' (1041) with group 'admin2 (1041)' ...
info: Creating home directory '/home/admin2' ...
info: Copying files from '/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for admin2
Enter the new value, or press ENTER for the default
Full Name []:
Room Number []:
Work Phone []:
Home Phone []:
Other []:
Is the information correct? [Y/n] y
info: Adding new user 'admin2' to supplemental / extra groups 'users' ...
```

Tester quelques comptes

```
ls: cannot access '/home/etudiant1/Maildir': Permission denied
lens@lens:~$ sudo ls -l /home/etudiant1/Maildir
total 12
drwx----- 2 etudiant1 etudiant1 4096 May  2 01:22 cur
drwx----- 2 etudiant1 etudiant1 4096 May  2 01:22 new
drwx----- 2 etudiant1 etudiant1 4096 May  2 01:22 tmp
lens@lens:~$ sudo ls -l /home/prof3/Maildir
total 12
drwx----- 2 prof3 prof3 4096 May  2 01:26 cur
drwx----- 2 prof3 prof3 4096 May  2 01:26 new
drwx----- 2 prof3 prof3 4096 May  2 01:26 tmp
lens@lens:~$ sudo ls -l /home/admin2/Maildir
total 12
drwx----- 2 admin2 admin2 4096 May  2 01:27 cur
drwx----- 2 admin2 admin2 4096 May  2 01:27 new
drwx----- 2 admin2 admin2 4096 May  2 01:27 tmp
lens@lens:~$ |
```

Installation de Roundcube (Webmail)

```
lens@lens: ~  
Get:3 http://ht.archive.ubuntu.com/ubuntu noble-updates/main amd64 unzip amd  
64 6.0-28ubuntu4.1 [174 kB]  
Fetched 334 kB in 9s (39.0 kB/s)  
Selecting previously unselected package php8.3-intl.  
(Reading database ... 176607 files and directories currently installed.)  
Preparing to unpack .../php8.3-intl_8.3.6-0ubuntu0.24.04.8_amd64.deb ...  
Unpacking php8.3-intl (8.3.6-0ubuntu0.24.04.8) ...  
Selecting previously unselected package php-intl.  
Preparing to unpack .../php-intl_2%3a8.3+93ubuntu2_all.deb ...  
Unpacking php-intl (2:8.3+93ubuntu2) ...  
Selecting previously unselected package unzip.  
Preparing to unpack .../unzip_6.0-28ubuntu4.1_amd64.deb ...  
Unpacking unzip (6.0-28ubuntu4.1) ...  
Setting up unzip (6.0-28ubuntu4.1) ...  
Setting up php8.3-intl (8.3.6-0ubuntu0.24.04.8) ...  
  
Creating config file /etc/php/8.3/mods-available/intl.ini with new version  
Setting up php-intl (2:8.3+93ubuntu2) ...  
Processing triggers for php8.3-fpm (8.3.6-0ubuntu0.24.04.8) ...  
NOTICE: Not enabling PHP 8.3 FPM by default.  
NOTICE: To enable PHP 8.3 FPM in Apache2 do:  
NOTICE: a2enmod proxy_fcgi setenvif  
NOTICE: a2enconf php8.3-fpm  
NOTICE: You are seeing this message because you have apache2 package install  
ed.  
Processing triggers for man-db (2.12.0-4build2) ...  
Processing triggers for libapache2-mod-php8.3 (8.3.6-0ubuntu0.24.04.8) ...  
Processing triggers for php8.3-cli (8.3.6-0ubuntu0.24.04.8) ...  
Scanning processes...  
Scanning linux images...  
  
Running kernel seems to be up-to-date.  
  
No services need to be restarted.  
  
No containers need to be restarted.  
  
No user sessions are running outdated binaries.  
  
No VM guests are running outdated hypervisor (qemu) binaries on this host.  
lens@lens:~$ |
```

Installation Roundcube

```
ens@lens:~$ sudo apt install roundcube roundcube-core roundcube-mysql -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
aspell aspell-en dbconfig-common dictionaries-common emacsen-common
enchant-2 fonts-glyphicons-halflings hunspell-en-us libaspell15
libenchant-2-2 libhunspell-1.7-0 libjs-bootstrap libjs-bootstrap4
libjs-codemirror libjs-jquery-minicolors libjs-jquery-ui
libjs-jstimedetect libjs-popover.js libjs-sizzle node-jquery
php-auth-sasl php-enchant php-getallheaders php-guzzlehttp-guzzle
php-guzzlehttp-promises php-guzzlehttp-psr7 php-mail-mime
php-masterminds-html5 php-net-sieve php-net-smtp php-net-socket php-pear
php-psr-http-client php-psr-http-factory php-psr-http-message
php-symfony-deprecation-contracts php8.3-enchant roundcube-skin-classic
roundcube-skin-larry
Suggested packages:
aspell-doc spellutils dbconfig-mysql | dbconfig-pgsql | dbconfig-sqlite3
| dbconfig-no-thanks wordlist hunspell openoffice.org-hunspell
| openoffice.org-core libenchant-2-voikko libjs-requirejs
libjs-jquery-ui-docs php-psr-log php-laminas-http-handlerrunner
roundcube-plugins roundcube-plugin-authres-status
roundcube-plugin-contextmenu roundcube-plugin-html5-notifier
roundcube-plugin-keyboard-shortcuts roundcube-plugin-listcommands
roundcube-plugin-message-highlight roundcube-plugin-sauserprefs
roundcube-plugin-thunderbird-labels php-crypt-gpg php-bacon-qr-code
php-roundcube-rtf-html-php php-net-ldap3 php-bjeavons-zxcvbn-php
mariadb-server | default-mysql-server
The following NEW packages will be installed:
aspell aspell-en dbconfig-common dictionaries-common emacsen-common
enchant-2 fonts-glyphicons-halflings hunspell-en-us libaspell15
libenchant-2-2 libhunspell-1.7-0 libjs-bootstrap libjs-bootstrap4
libjs-codemirror libjs-jquery-minicolors libjs-jquery-ui
libjs-jstimedetect libjs-popover.js libjs-sizzle node-jquery
php-auth-sasl php-enchant php-getallheaders php-guzzlehttp-guzzle
php-guzzlehttp-promises php-guzzlehttp-psr7 php-mail-mime
php-masterminds-html5 php-net-sieve php-net-smtp php-net-socket php-pear
php-psr-http-client php-psr-http-factory php-psr-http-message
```

Configuring roundcube-core

The roundcube package must have a database installed and configured before it can be used. This can be optionally handled with `dbconfig-common`.

If you are an advanced database administrator and know that you want to perform this configuration manually, or if your database has already been installed and configured, you should refuse this option. Details on what needs to be done should most likely be provided in `/usr/share/doc/roundcube`.

Otherwise, you should probably choose this option.

Configure database for roundcube with `dbconfig-common`?

<Yes>

<No>

Créer la configuration Nginx



```
lens@lens: ~  
GNU nano 7.2 /etc/nginx/sites-available/roundcube *  
server {  
    listen 80;  
    server_name _;  
  
    root /usr/share;  
    index index.php;  
  
    location /roundcube {  
        index index.php;  
        try_files $uri $uri/ /roundcube/index.php?$query_string;  
    }  
  
    location ~ \.php$ {  
        include snippets/fastcgi-php.conf;  
        fastcgi_pass unix:/run/php/php8.4-fpm.sock;  
    }  
}
```

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify

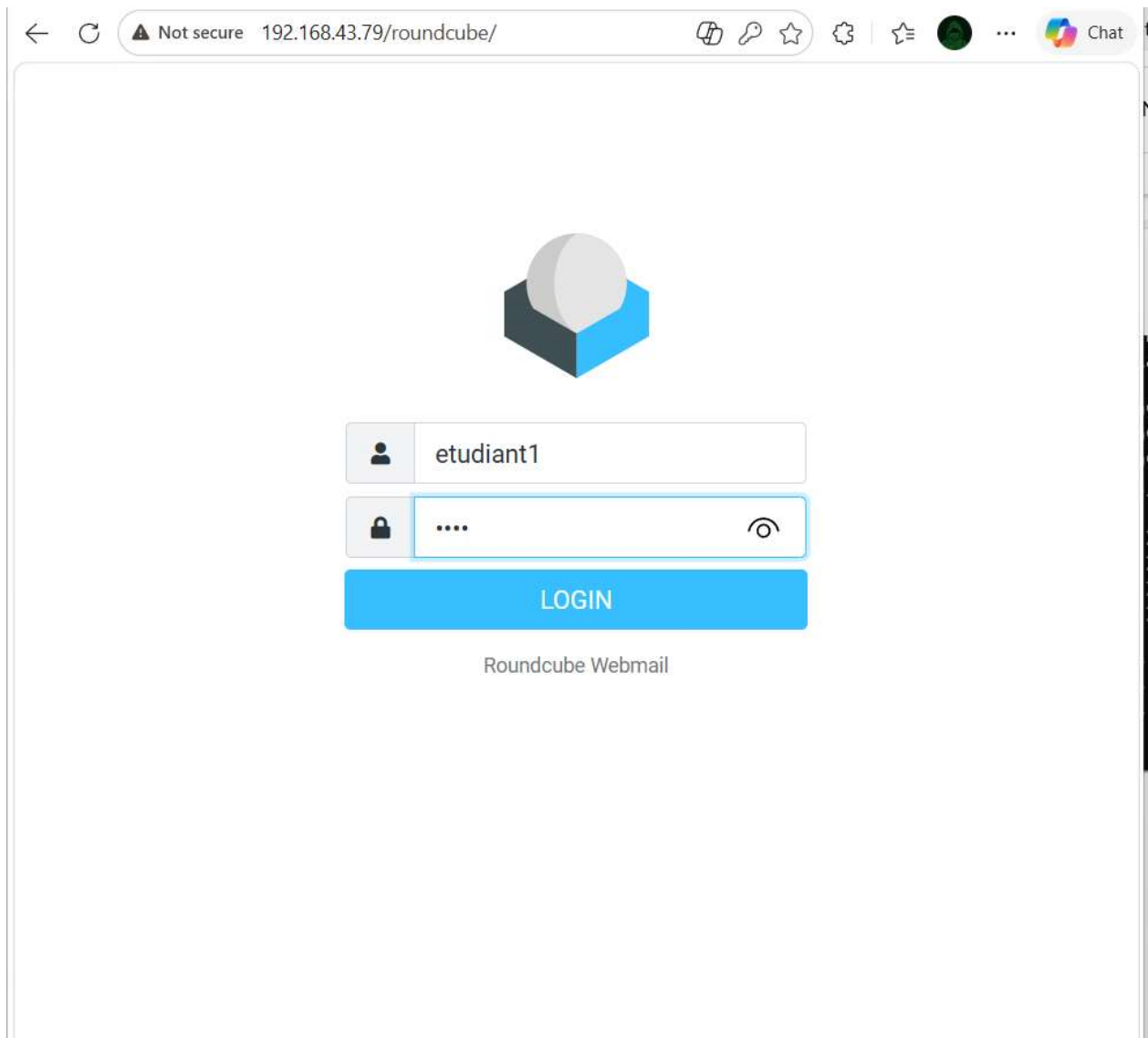
Activation du site et Test de la configuration


```
lens@lens:~$ sudo nano /etc/nginx/sites-available/roundcube
lens@lens:~$ sudo ln -s /etc/nginx/sites-available/roundcube /etc/nginx/sites-enabled/
lens@lens:~$ sudo rm /etc/nginx/sites-enabled/default
lens@lens:~$ sudo nginx -t
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
lens@lens:~$ |
```

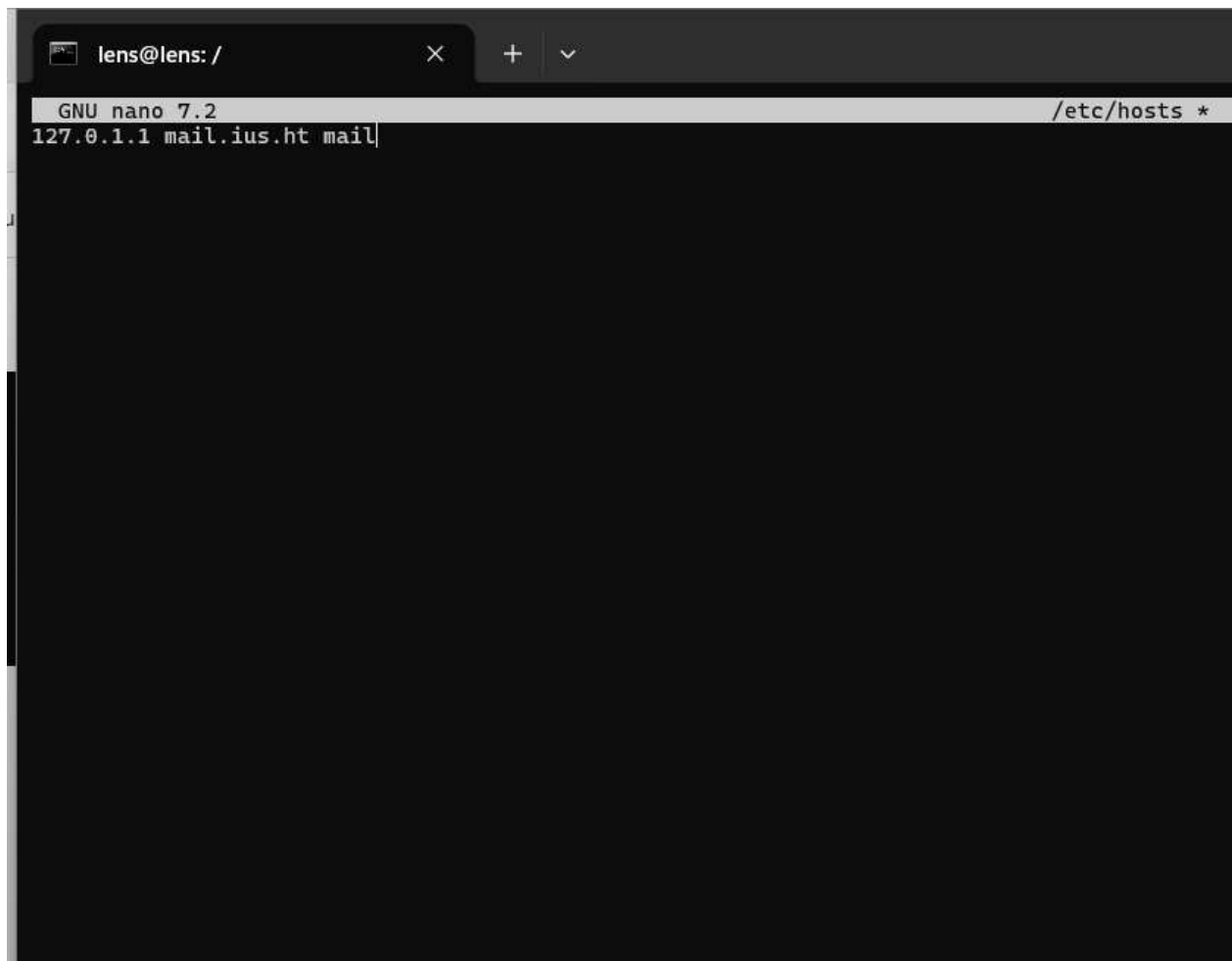
Redémarrage des services

```
lens@lens:~$ sudo systemctl restart php8.3-fpm
lens@lens:~$ sudo systemctl restart nginx
lens@lens:~$ |
```

Accès

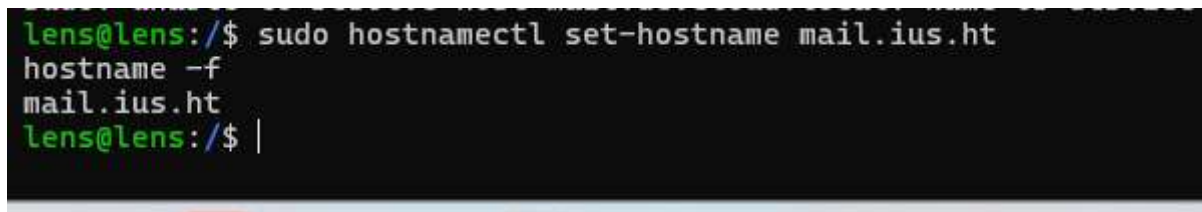


Modifier /etc/hosts



The screenshot shows a terminal window with a dark background. At the top, a tab is labeled 'lens@lens: /'. Below the tab, the text 'GNU nano 7.2' is visible on the left and '/etc/hosts *' on the right. The main content of the terminal shows the first line of the /etc/hosts file: '127.0.1.1 mail.ius.ht mail'. The cursor is positioned at the end of this line.

```
lens@lens: /  
GNU nano 7.2 /etc/hosts *  
127.0.1.1 mail.ius.ht mail
```



The screenshot shows a terminal window with a dark background. The prompt is 'lens@lens:/\$'. The command 'sudo hostnamectl set-hostname mail.ius.ht' has been entered. The output shows 'hostname -f' followed by 'mail.ius.ht' on the next line. The prompt is now 'lens@lens:/\$' with a cursor at the end.

```
lens@lens:/$ sudo hostnamectl set-hostname mail.ius.ht  
hostname -f  
mail.ius.ht  
lens@lens:/$
```

Tests fonctionnels

```

lens@lens:/$ su - admin
ls -la ~
su: user admin does not exist or the user entry does not contain all the required fields
total 316424
drwxr-x--- 15 lens lens      4096 May  3 21:21 .
drwxr-xr-x 46 root root      4096 May  2 01:14 ..
drwxrwxr-x  3 lens lens      4096 Mar 24 23:14 .git
drwxrwxr-x  5 lens lens      4096 Apr 19 12:58 .ansible
-rw-rw-r--  1 lens lens      4498 May  2 02:00 backup_prod.log
-rw-----  1 lens lens    18606 May  2 18:10 .bash_history
-rw-r--r--  1 lens lens       220 Mar 31 2024 .bash_logout
-rw-r--r--  1 lens lens      3771 Mar 31 2024 .bashrc
drwx-----  2 lens lens      4096 Mar  8 17:23 .cache
drwx-----  4 lens lens      4096 Mar 21 18:27 .config
-rw-----  1 lens lens       399 Mar 15 18:04 devcloud
-rw-r--r--  1 lens lens       91 Mar 15 18:04 devcloud.pub
-rw-r--r--  1 root root         1 Mar 29 20:49 g
-rw-rw-r--  1 lens lens       96 Apr 19 12:56 inventory.ini
-rw-----  1 lens lens       20 May  3 21:21 .lessht
drwxrwxr-x  3 lens lens      4096 Mar 15 16:44 .local
drwxrwxr-x  5 lens lens      4096 May  2 03:09 Maildir
drwxr-xr-x  2 lens lens      4096 Oct 25 2025 node_exporter-1.10.2.linux-amd64
-rw-rw-r--  1 lens lens 11686705 Oct 25 2025 node_exporter-1.10.2.linux-amd64.tar.gz
-rw-r--r--  1 lens lens       807 Mar 31 2024 .profile
drwxr-xr-x  3 lens lens      4096 Apr 26 21:38 prometheus-3.11.0.linux-amd64
-rw-rw-r--  1 lens lens 150167744 Apr  2 11:05 prometheus-3.11.0.linux-amd64.tar.gz
-rw-rw-r--  1 lens lens 11794169 Apr 26 19:19 prometheus-3.11.0.linux-amd64.tar.gz.1
-rw-rw-r--  1 lens lens 150167744 Apr  2 11:05 prometheus-3.11.0.linux-amd64.tar.gz.2
drwxrwxr-x  5 lens lens      4096 Apr 19 12:59 roles
-rwxrwxr-x  1 lens lens       760 Mar 15 20:46 sauvegarde_distante.sh
-rwxrwxr-x  1 lens lens       569 Mar 15 16:55 sauvegarde.sh
-rw-rw-r--  1 lens lens         0 Mar 15 16:56 .selected_editor
drwxrwxr-x  4 lens lens      4096 Apr 19 21:02 site_jinja
-rw-rw-r--  1 lens lens      149 Apr 19 14:19 site.yml
drwx-----  2 lens lens      4096 Apr 19 12:34 .ssh
drwxrwxr-x  2 lens lens      4096 Apr 14 00:41 ssl_certificate
-rw-r--r--  1 lens lens         0 Mar  8 17:30 .sudo_as_admin_successful
-rw-r--r--  1 lens lens      8192 Apr  5 16:51 'systemctl status apache2'
-rw-rw-r--  1 lens lens      1457 Apr 19 14:13 td.html
drwxrwxr-x  2 lens lens      4096 Apr 19 12:35 tp-ansible
-rw-rw-r--  1 lens lens       205 Apr 26 21:52 .wget-hsts
-rw-rw-r--  1 lens lens      4555 Apr 26 19:43 wget-log
-rw-----  1 lens lens       399 Mar 15 18:05 yes
-rw-r--r--  1 lens lens       91 Mar 15 18:05 yes.pub

```

Alias de groupe

```
lens@lens: /  
GNU nano 7.2 /etc/aliases *  
# See man 5 aliases for format  
postmaster:    root  
itteam: etudiant1,prof1,admin1|
```

```
lens@lens:/$ echo "Réunion à 14h en salle serveur" | mail -s "Information équipe" admin dev suppo  
lens@lens:/$ sudo nano /etc/aliases  
lens@lens:/$ sudo newaliases  
lens@lens:/$ echo "Maintenance prévue ce soir à 22h" | mail -s "Alerte maintenance" itteam  
lens@lens:/$ |
```

Vérification

```
etudiant1@mail:~$ mail  
Mail version 8.1.2 01/15/2001. Type ? for help.  
"/var/mail/etudiant1": 1 message 1 new  
>N 1 lens@mail.ius.ht Sun May 3 22:28 17/548 Alerte maintenance  
&
```

Sécurisation

```

lens@lens:/$ sudo ufw allow 25
Rules updated
Rules updated (v6)
lens@lens:/$ sudo ufw allow 143
Skipping adding existing rule
Skipping adding existing rule (v6)
lens@lens:/$ sudo ufw allow 993
Skipping adding existing rule
Skipping adding existing rule (v6)
lens@lens:/$ sudo ufw allow 80
Skipping adding existing rule
Skipping adding existing rule (v6)
lens@lens:/$ sudo ufw enable
Command may disrupt existing ssh connections. Proceed with operation (y|n)? y
Firewall is active and enabled on system startup
lens@lens:/$
lens@lens:/$ |

```

Analyse des logs

```

lens@lens:/$
lens@lens:/$ sudo tail -f /var/log/mail.log
2026-05-03T22:28:55.639811+00:00 lens postfix/cleanup[21076]: 994252266: message-id=<202605032228
2026-05-03T22:28:55.658355+00:00 lens postfix/qmgr[1245]: 994252266: from=<lens@mail.ius.ht>, siz
2026-05-03T22:28:55.663881+00:00 lens postfix/local[21078]: warning: dict_nis_init: NIS domain na
2026-05-03T22:28:55.670721+00:00 lens postfix/local[21078]: 994252266: to=<etudiant1@mail.ius.ht>
/0.01, dsn=2.0.0, status=sent (delivered to mailbox)
2026-05-03T22:28:55.822388+00:00 lens postfix/local[21078]: warning: host not found: localhost
2026-05-03T22:28:55.827796+00:00 lens postfix/local[21078]: 994252266: to=<prof1@mail.ius.ht>, or
6, dsn=2.0.0, status=sent (delivered to mailbox)
2026-05-03T22:28:55.995566+00:00 lens postfix/local[21078]: warning: host not found: localhost
2026-05-03T22:28:56.001292+00:00 lens postfix/local[21078]: 994252266: to=<admin1@mail.ius.ht>, o
34, dsn=2.0.0, status=sent (delivered to mailbox)
2026-05-03T22:28:56.175516+00:00 lens postfix/local[21078]: warning: host not found: localhost
2026-05-03T22:28:56.186035+00:00 lens postfix/qmgr[1245]: 994252266: removed
^Z
[48]+  Stopped                  sudo tail -f /var/log/mail.log
lens@lens:/$ |

```


Cocclusion.

Ce TD m'a permis de savoir comment configurer la réception IMAP / POP3, déployer un webmail professionnel, sécuriser les accès et documenter une architecture mail complete. donc la tache a été réussie.